nature sustainability

Article

https://doi.org/10.1038/s41893-025-01764-w

Electrosynthesis of ethylene from syngas

Received: 6 January 2025

Accepted: 19 December 2025

Published online: 10 February 2026

Check for updates

Feng Li[1,9], Zunmin Guo[1,9], Yu Yan[2,9], Qiyou Wang[1,9], Yongxiang Liang[2], Lian Duan[3], Loann Bonnenfant[1,4], Zahra Azimi Dijvejin[1], Min Liu[1], Yurou Celine Xiao[1], Ziyu Zhao[5], Jieyuan Liu[1], Andrija Stepanovic[1], Yuxuan Gao[1], Qian Sun[1], Hafiz Ghulam Abbas[2], Mengyang Fan[1], Jianan Erick Huang[2], Sungjin Park[2], Stuart M. Holmes[5], Yong Zhao[6], Yi Xu[1], Cao-Thang Dinh[7], Ziyun Wang[8] ✉, Rui Kai Miao[1] ✉ & David Sinton[1] ✉

The electrochemical carbon monoxide reduction (COR) reaction offers a carbon- and energy-efficient pathway to ethylene. However, present systems typically rely on pure CO, making the process costly and difficult to scale. An economically feasible alternative is to use syngas—the predominant industrial CO source—produced globally at quantities approximately 250 times higher than pure CO, but containing hydrogen and other components. Therefore, for a syngas-based COR system, achieving high single-pass conversion efficiency (SPCE) is essential, as high gas flow rates introduce excess hydrogen, complicating and increasing the cost of downstream product separation. Nevertheless, optimizing SPCE leads to hydrogen dilution and insufficient CO transport to the copper cathode, which in turn decreases ethylene Faradaic efficiency (FE). To overcome this trade-off, guided by Raman spectroscopy and density functional theory, here we develop a COOH-functionalized active carbon material which captures CO from syngas, strengthens CO adsorption on the copper cathode and thereby enhances ethylene FE even under CO-diluted conditions. This design enables an ethylene FE of 72% alongside a CO SPCE of 73% in a simulated syngas containing an H_2/CO ratio of 2/1—outperforming reported systems operated with pure CO feed and offering a more scalable, resource-efficient route to sustainable ethylene.

Electrochemical reduction of carbon dioxide (CO_2R) using renewable electricity, if achieved with high efficiency, offers a promising route to decarbonize chemical production[1,2]. CO_2R in alkaline and neutral electrolytes enables fast kinetics and high selectivity towards multicarbon (C_{2+}) products, such as ethylene (C_2H_4) (ref. 3). However, in these electrolytes, CO_2 reacts with hydroxide to form carbonate, leading to CO_2 losses that require energy-intensive regeneration—more than twice the energy needed for electrolysis[4]. Acidic CO_2R, while enhancing CO_2 use, suffers from the competitive hydrogen evolution reaction (HER), limiting C_2H_4 energy efficiency (EE) to less than 15% (ref. 5).

Cascade CO_2 electrolysis, dividing CO_2–CO–C_2H_4 into two separate steps, offers a more energy- and carbon-efficient route to C_2H_4 (Fig. 1a)[4,6–16]. Among the technologies that convert CO_2 to CO, high-temperature solid oxide electrolysers (SOECs) are the most efficient, achieving EE exceeding 80% (ref. 6). However, SOECs are prone to carbon deposition (also known as 'coking'), posing stability

[1]Department of Mechanical and Industrial Engineering, University of Toronto, Toronto, Ontario, Canada. [2]The Edward S. Rogers Department of Electrical and Computer Engineering, University of Toronto, Toronto, Ontario, Canada. [3]School of Mining and Petroleum Engineering, Department of Civil and Environmental Engineering, University of Alberta, Edmonton, Alberta, Canada. [4]New Energies & Electricity R&D unit, TotalEnergies SE, Paris-Saclay, France. [5]Department of Chemical Engineering, University of Manchester, Manchester, UK. [6]Global Innovative Centre for Advanced Nanomaterials, School of Science, University of Newcastle, Newcastle, New South Wales, Australia. [7]Department of Chemical Engineering, Queen's University, Kingston, Ontario, Canada. [8]School of Chemical Sciences, University of Auckland, Auckland, New Zealand. [9]These authors contributed equally: Feng Li, Zunmin Guo, Yu Yan, Qiyou Wang. ✉e-mail: ziyun.wang@auckland.ac.nz; ruikai.miao@mail.utoronto.ca; sinton@mie.utoronto.ca

https://doi.org/10.1038/s41893-025-01764-w

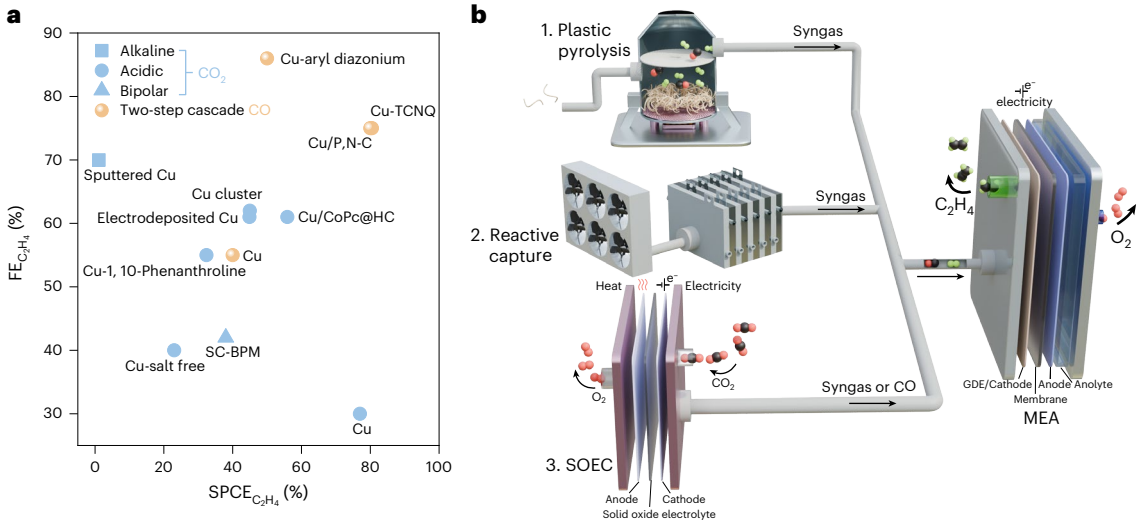

Fig. 1 | Faradaic and carbon efficiency of C_2H_4 electrosynthesis in different electrolysers. a, FE and SPCE performance assessment of literature benchmark CO_2R to ethylene systems are reported in refs. 4,8–16. CoPc@HC, hollow carbon-supported cobalt phthalocyanine; SC-BPM, stationary catholyte-based bipolar membrane; TCNQ, 7,7,8,8-tetracyanoquinodimethane; P,N-C, phosphorus and nitrogen co-doped carbon. **b**, Schematics of routes to ethylene electrosynthesis via syngas produced from plastic pyrolysis, reactive capture or SOEC upstream. GDE, gas diffusion electrode.

challenges[17,18]. Coking arises from the thermal decomposition or disproportionation of CO ($CO \rightarrow C + CO_2$) under high CO concentrations, leading to the accumulation of carbon deposits that block active sites and degrade SOEC performance. Although these challenges may be surmountable, the global market for pure CO remains relatively small (~1 million t yr^{-1})[19] and technology developers are thus motivated to serve larger markets by generating hydrogen or syngas—a mixture of CO and H_2 (ref. 20).

In contrast to pure CO, syngas is abundant, providing an alternative feed for CO electroreduction. Produced today predominantly via steam-methane reforming, syngas serves as the intermediates for long-chain chemicals, with global production exceeding 250 million t yr^{-1}(refs. 21,22) (Fig. 1b). Motivated by this scale, there is growing interest in low-carbon routes to produce syngas. SOECs configured for syngas production benefit from the H_2 dilution, which suppresses coking[23] and enables stable, long-term operation at scale[24]. Reactive capture—which integrates CO_2 capture and conversion—offers another route to syngas[25–27]. Additionally, emerging technologies such as plastic pyrolysis can convert hard-to-recycle plastics into syngas[28,29].

The H_2 in the syngas is not a reactant at the cathode of CO electrolysers and separating H_2 from CO before electrolysis is costly, with pure CO costing over ten times more than syngas[30]. Additionally, HER at the cathode produces H_2 downstream and thus application of conventional CO electrolysers would require H_2 separation both upstream and downstream of electrolysis. These challenges motivate the direct reduction of syngas—with CO concentrations varying from 10% to 50%—into products at a high rate of efficiency (Fig. 1b).

In this work, we identify CO mass-transport limitations as the fundamental bottleneck preventing high single-pass conversion efficiency (SPCE) in syngas-fed CO electroreduction systems. To overcome this challenge, we design a CO-shuttling medium based on COOH-functionalized carbon materials that selectively captures CO from syngas and reinforces its adsorption on Cu, thereby delivering CO directly to catalytic sites. This approach ensures sufficient CO supply even under SPCE-optimized, CO-diluted conditions. By integrating this shuttling mechanism, we achieve a C_2H_4 Faradaic efficiency (FE) of 72% and a CO SPCE of 73% in syngas with an H_2/CO ratio of 2/1—surpassing the performance of systems operating on pure CO feed.

Results

Syngas compositions, defined by their H_2/CO ratios, vary according to synthesis conditions[31,32]. We evaluated carbon monoxide reduction (COR) performance using Cu electrode derived from two-dimensional copper oxide (CuO) nanosheets (Methods; Supplementary Figs. 1 and 2) in a membrane electrode assembly (MEA; Supplementary Fig. 3) under syngas with H_2/CO molar ratios of 1/1, 2/1 and 3/1. We selected CuO-derived Cu nanosheets over commercial Cu nanoparticles because of their higher C_2H_4 FE (Supplementary Fig. 4). The higher C_2H_4 selectivity observed for copper nanosheets relative to commercial copper nanospheres may reflect a greater density of catalytically competent sites formed during the in situ reduction of CuO nanosheets used in CO_2 reduction[33–35], along with possible influences of nanoscale morphology and the local interfacial microenvironment. However, these factors warrant future investigation, as design rules can differ between CO_2 and CO electroreduction.

Factors affecting syngas reduction

We obtained C_2H_4 FE ranging from 50% to 55% with syngas at 200 mA cm^{-2}, surpassing the C_2H_4 FE observed for pure CO feed under similar conditions (47%; Fig. 2a). We attributed this improvement to CO dilution by H_2, which constrains CO coverage at the catalyst surface, stabilizing key intermediates for ethylene formation[36]. We also considered whether the presence of H_2 could shift the equilibrium of the competing HER towards reactants, reducing H_2 production and indirectly enhancing C_2H_4 FE. However, linear sweep voltammetry (LSV) under N_2 and H_2 conditions showed similar trends, indicating that H_2 acts only as a diluent without substantially impacting HER kinetics (Supplementary Fig. 5). We also assessed the influence of electrolyte composition and found that 1 M LiOH provided the highest FE towards C_2H_4 (Supplementary Fig. 6).

Achieving high SPCE is important for all electrolysers but particularly for syngas-based COR systems, as H_2 is not reacted and high inlet flow rates introduce excess H_2 that increase all downstream separation costs (Fig. 2b,c). Thus, we constrained syngas input into the cell and focused on more CO-rich streams prevalent in industry, with H_2/CO ratios of 1/1 and 2/1 (refs. 32,37,38). Reducing input flow rates to 2 sccm (standard cubic centimeters per minute) increased SPCE to 86% and 89% for an H_2/CO ratio of 1/1 and 2/1, respectively. However, the improvement came at the cost of C_2H_4 FE, which decreased to 42% and

https://doi.org/10.1038/s41893-025-01764-w

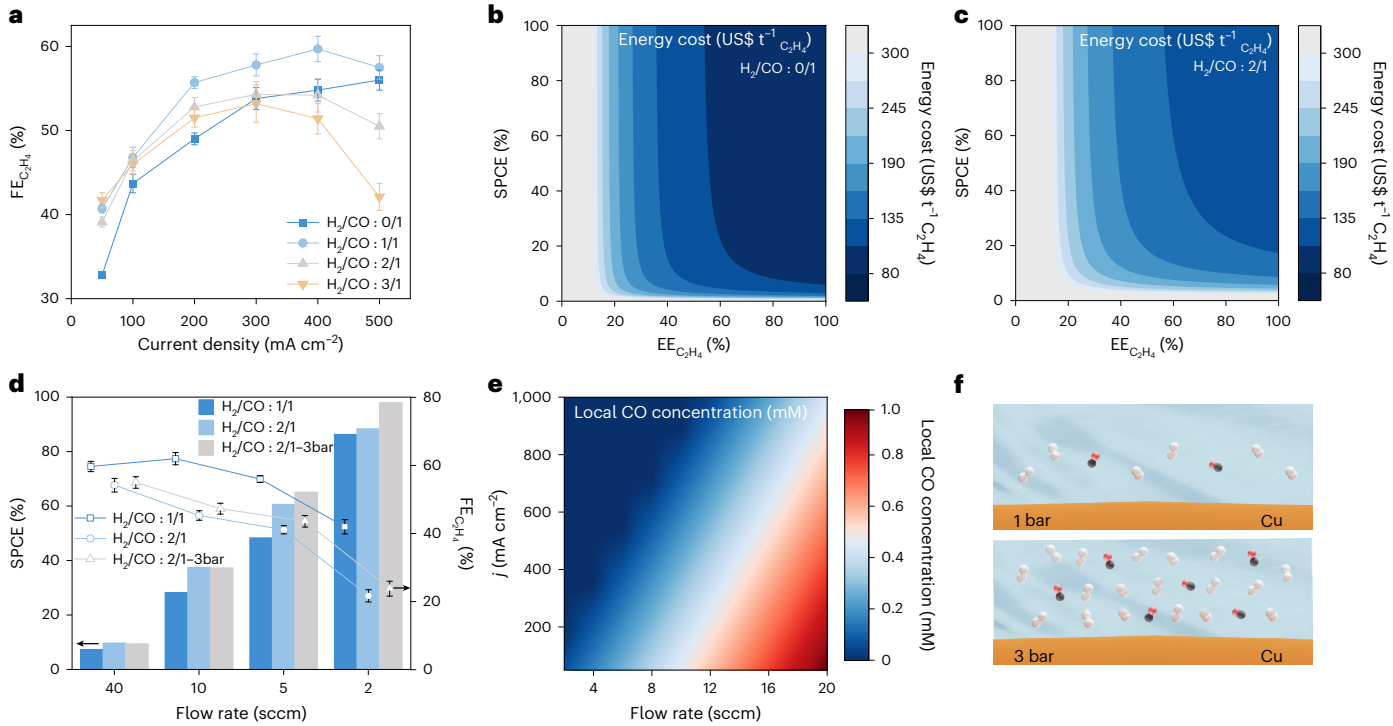

Fig. 2 | Key parameters affecting performance of COR from syngas towards C_2H_4. **a**, C_2H_4 FE during COR from syngas with various H_2/CO ratios at 40 sccm in 1 M LiOH. **b**,**c**, C_2H_4 energy cost as the function of C_2H_4 SPCE and EE during COR from pure CO (**b**) and H_2/CO ratios of 2/1 (**c**). Colour scales in **b** and **c** represent the energy cost per ton of C_2H_4 (US$ t^{-1} C_2H_4). **d**, Effect of flow rate and pressure on C_2H_4 SPCE and FE. **e**, Dependence of local CO concentration on reaction current density and flow rate. Colour scale represents local CO concentration (mM). **f**, Schematics of local species distribution at 1 bar (top panel) and 3 bar (bottom panel). Where error bars are shown, values are means and error bars indicate s.d. (n = 3 replicates).

22% for an H_2/CO ratio of 1/1 and 2/1 (Fig. 2d), respectively. Reaction–diffusion modelling (Fig. 2e) indicated that CO becomes mass-transport limited as the flow rate decreases, favouring HER. To validate this, we further evaluated COR from an H_2/CO ratio of 1/1 at 2 sccm across different applied current densities. These results show that the H_2 FE remains above 40%, while the total FE of COR products stays below 60% (Supplementary Fig. 8a)—a trend similar to that observed for an H_2/CO ratio of 1/3 at 3 sccm (Supplementary Fig. 8b)—confirming that the local CO availability becomes limiting under these conditions. To explore the reason, we performed the COR using pure CO at same molar feed (1 sccm), where H_2 FE decreases to below 20% and total FE of COR products increases to ~80% within 100–300 mA cm^{-2} range (Supplementary Fig. 8c). These results indicate that the H_2 dilution effect is a primary factor limiting CO mass transport under high-SPCE syngas conditions.

We then attempted to increase CO mass transport by pressurizing the reactant stream. However, operating at 3 bar resulted in marginal improvements in C_2H_4 FE and SPCE (Fig. 2d). This modest gain suggests that pressurization increases local availability of both CO and H_2 near the catalyst (Fig. 2f), without substantially changing the CO/H_2 ratio and that bulk pressure control alone cannot resolve CO transport limitations in syngas-fed COR. On the basis of this finding, we hypothesize that selectivity increasing local CO availability at catalytic active sites, relative to H_2, may help preserve C_2H_4 FE under high-SPCE conditions.

Concentrating local CO through carbon material modification

We explored using a medium to selectively capture CO and enhance its concentration near catalytic sites relative to H_2 (Fig. 3a)[11,39,40]. Porous carbon materials are promising candidates for this purpose because of their high CO affinity and cost-effectiveness[41,42]. To test this concept, we modified Cu electrodes with carbon materials of varying curvatures, including graphene oxide (Cu/GO), carbon nanotube (Cu/CNT) and carbon black (Cu/C) (Supplementary Fig. 9) and evaluated their performance at an H_2/CO ratio of 1/1. The modified Cu catalysts exhibited improved C_2H_4 FE compared with bare Cu, with Cu/C achieving the highest C_2H_4 FE, exceeding 55% at 200 mA cm^{-2} (Fig. 3b). Operando Raman spectroscopy revealed a more positive onset potential for CO adsorption on Cu/C (−1.4 V versus Ag/AgCl), compared with bare Cu (−1.6 V versus Ag/AgCl) (Supplementary Figs. 10 and 11), indicating an increased CO concentration on the Cu surface facilitated by carbon black.

However, we observed a decline in C_2H_4 FE on Cu/C when operating beyond 200 mA cm^{-2} (Fig. 3b). To investigate the cause, we returned to Raman spectroscopy and identified weakened CO adsorption on the Cu active sites, evidenced by the blue shift of C–O stretching peak at 1,900–2,200 cm^{-1} (Fig. 3c)[43]. This suggests that while carbon black effectively enhances CO availability near the Cu surface, it may inadvertently alter the CO–Cu interactions, reducing CO adsorption strength (Fig. 3a). This finding prompted us to further explore materials capable of simultaneously capturing CO and maintaining strong CO adsorption on the catalytic surface.

Enhancing CO adsorption through surface functionalization

We considered the surface functionalization of carbon black to enhance CO adsorption strength on Cu active sites, facilitating the shuttling of CO from the bulk electrolyte towards the Cu surface. Using density functional theory (DFT) calculations, we investigated two key aspects: (1) CO adsorption on Cu surfaces modified with various functionalized carbon blacks, including N, P and COOH (Fig. 3d,e)[44–46] and (2) the intrinsic CO adsorption energies of the support themselves, indicating their ability to locally concentrate CO near the Cu surface (Fig. 3f,g).

Among the studied materials, COOH-functionalized carbon edge (Cu/C_{edge}–COOH) showed the strongest CO adsorption on Cu (−1.21 eV), exceeding bare Cu (−1.08 eV) (Fig. 3e) and other functionalized variants such as Cu surfaces modified by N-doped carbon edge (Cu/C_{edge}–N,

https://doi.org/10.1038/s41893-025-01764-w

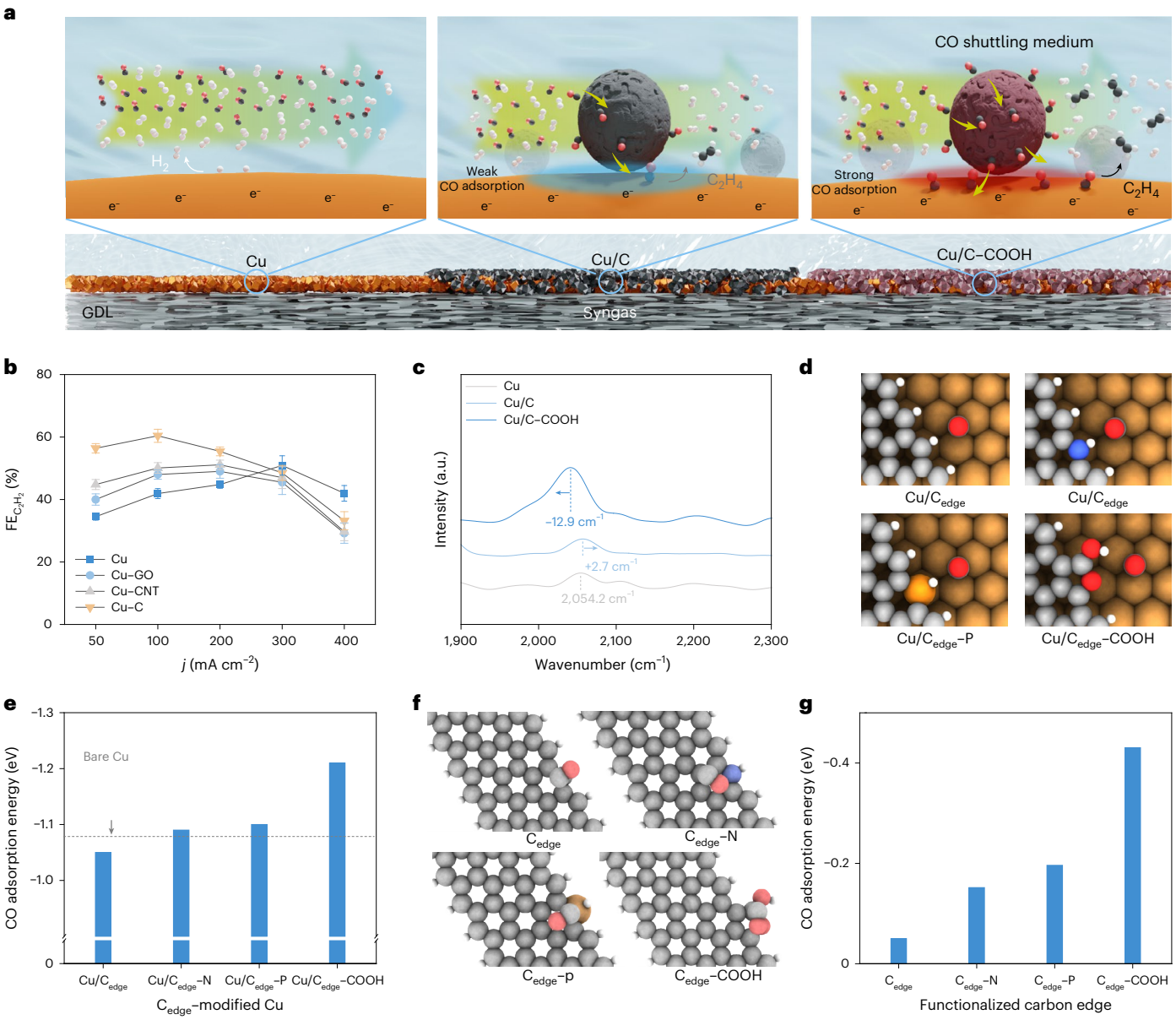

Fig. 3 | Mechanism study of carbon modification on Cu for CO adsorption. **a**, Schematics of the local environment on Cu, Cu/C and Cu/C–COOH during COR from syngas. GDL, gas diffusion layer. **b**, C_2H_4 FE from H_2/CO ratio of 1/1 on pristine Cu, GO-, CNT- and C-modified Cu at 2 sccm. **c**, Operando Raman test of pristine Cu, Cu modified by C and C–COOH under onset potentials (versus Ag/AgCl reference electrode filled with saturated aqueous KCl) in electrolytes of 1 M LiOH. **d**,**e**, Optimized geometries of CO adsorption (**d**) and their corresponding adsorption energies on Cu surfaces modified by functionalized carbons (**e**). **f**,**g**, Optimized geometries of CO adsorption (**f**) and their corresponding adsorption energies on functionalized carbon edges (**g**). Where error bars are shown, values are means and error bars indicate s.d. ($n = 3$ replicates).

−1.09 eV), P-doped carbon edge (Cu/C_{edge}–P, −1.10 eV) and unmodified carbon edge (−1.05 eV). This enhancement also persisted across a range of CO coverage conditions (Supplementary Fig. 13). C_{edge}–COOH (−0.43 eV) also exhibited the highest CO adsorption energy as a support material (−0.43 eV), outperforming bare carbon black (−0.05 eV), C_{edge}–N (−0.15 eV) and C_{edge}–P–C (−0.19 eV) (Fig. 3g), as well as oxygen-functionalized groups such as –OH (−0.12 eV) and –C = O (−0.13 eV) (Supplementary Fig. 14).

Considering the randomness in functional group distribution and the structural complexity of carbon materials, we extended our analysis beyond functionalized carbon edges (Fig. 3d,f) to include functionalized carbon surfaces (Supplementary Figs. 15–26) and amorphous carbons (Supplementary Tables 6–21). The functional group densities used in these models (~5.5–6.2% for carbon surfaces and ~4.6–5.7% for the amorphous carbons) closely match the experimental values of ~5.2–6.1% obtained from X-ray photoelectron spectroscopy (XPS) analysis (Supplementary Fig. 27). Notably, COOH-functionalized carbon surfaces, amorphous carbons and their corresponding modified-Cu surfaces exhibited the strongest enhancement in CO adsorption compared with bare and N- or P-functionalized carbons (Supplementary Figs. 28, 29 and 31–33), consistently observed across a Cu–COOH distance range of 2.2–4.3 Å.

Together, these results identify C–COOH as the most promising candidate, offering dual benefits: enhanced CO retention on the carbon support and strengthened CO adsorption on the Cu surface.

Experimentally, we synthesized C–COOH with CO uptake ability by treating the active carbon with nitric acid (Methods). Scanning electron microscopy (SEM) showed that the C–COOH retained the structure

https://doi.org/10.1038/s41893-025-01764-w

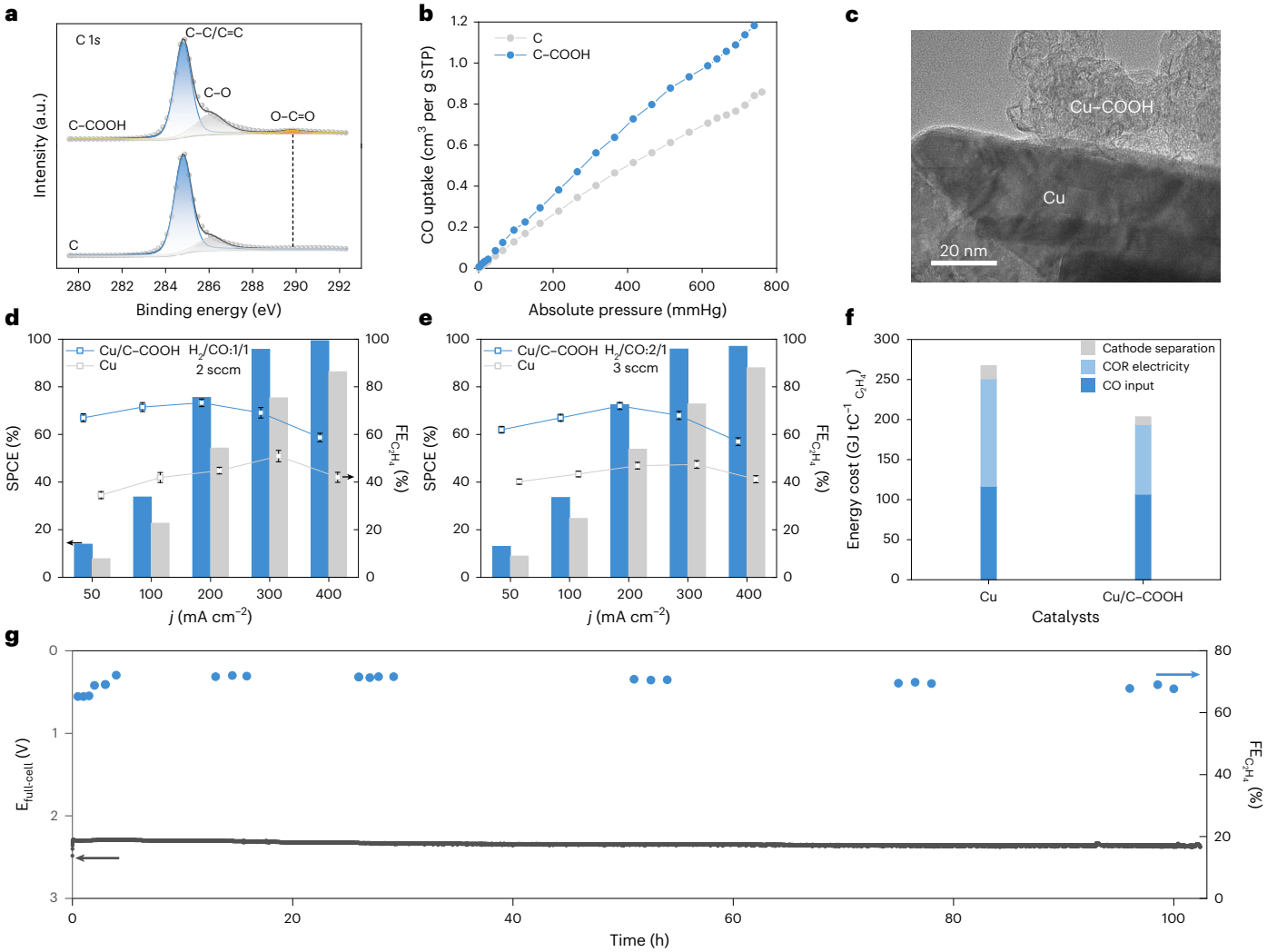

Fig. 4 | Performance of COR from syngas of Cu/C–COOH electrode. a, XPS C 1*s* spectra for C and C–COOH. **b**, Adsorption isotherms of CO for C and C–COOH at 298 K. **c**, TEM image of Cu/C–COOH. **d**,**e**, C_2H_4 FE and SPCE of the pristine Cu and Cu/C–COOH for COR from H_2/CO ratio of 1/1 at 2 sccm (**d**) and H_2/CO ratio of 2/1 at 3 sccm (**e**). **f**, Energy cost over Cu/C–COOH and bare Cu for H_2/CO ratio of 2/1 at 3 sccm based on the cascade CO_2R system. **g**, Stability of the Cu/C–COOH for COR from H_2/CO ratio of 2/1 at a constant current density of 200 mA cm^{-2} and flow rate of 3 sccm, with the corresponding C_2H_4 FE. Where error bars are shown, values are means and error bars indicate s.d. (n = 3 replicates).

of the original carbon black (Supplementary Fig. 34). XPS (Fig. 4a) and energy-dispersive X-ray spectroscopy (EDS) elemental mapping (Supplementary Fig. 35) verified the successful incorporation of COOH functional groups. Gas adsorption isotherms measurements revealed that C–COOH exhibited higher CO uptake than bare carbon (Fig. 4b) while maintaining a comparable H_2 uptake (Supplementary Fig. 36), suggesting that the COOH functional groups selectively enhance CO adsorption, in agreement with DFT calculations (Fig. 3g).

Syngas reduction on functionalized Cu/C electrodes

We then assembled C–COOH with Cu to form the Cu/C–COOH electrode (Fig. 4c). N_2O titration experiments indicated a fractional exposure of Cu of ~0.19% for Cu/C–COOH electrode, similar to the ~0.18% of bare Cu electrode (Supplementary Fig. 37 and Supplementary Note 1)[47]. Operando Raman measurements revealed enhanced CO adsorption on Cu/C–COOH, compared with Cu/C, as evidenced by a red shift of C–O stretching peak at 1,900–2,200 cm^{-1} (Fig. 3c)[48]. In addition, Cu/C–COOH exhibited the lowest onset potential for CO adsorption, compared with −1.2 V versus −1.4 V for Cu/C and −1.6 V for bare Cu (Supplementary Figs. 10–12), suggesting a simultaneous enhancement in both CO availability and CO adsorption[49] at the catalytic interface.

We evaluated the Cu/C–COOH electrode in syngas at high-SPCE conditions. At an H_2/CO ratio of 1/1, the Cu/C–COOH electrode demonstrated a C_2H_4 FE of 73% and H_2 FE of 11% with a SPCE of 76% at 200 mA cm^{-2} (Fig. 4d and Supplementary Fig. 40a). In contrast, Cu/C–N and Cu/C–P showed lower C_2H_4 FE of 51% and 67%, respectively (Supplementary Fig. 41), in agreement with DFT calculations (Fig. 3e,g). To assess performance under more CO-dilute conditions (H_2/CO ratio of 2/1), we adjusted the total input flow rate to maintain the same CO partial flow rate as the 1/1 case. The Cu/C–COOH electrode continued to work well even at a lower CO content with an H_2/CO ratio of 2/1, achieving a C_2H_4 FE of 72% and H_2 FE of 13% with a SPCE of 73% at an input flow rate of 3 sccm (Fig. 4e and Supplementary Fig. 40b). However, when the total flow was reduced to 2 sccm, the C_2H_4 FE decreased substantially because of insufficient CO supply and excess H_2 (Supplementary Fig. 42).

Long-term stability measurement

To evaluate the stability of Cu/C–COOH electrode, we replaced commercial carbon paper with a custom-made carbon paper containing high PTFE loading (Methods) to enhance hydrophobicity. We also substituted IrO_2 anode catalyst with nickel–iron–boron (Ni–Fe–B) to reduce

the full-cell voltage. The gas product distribution was unchanged with these optimizations (Supplementary Fig. 44). We achieved a full-cell voltage of 2.3 V at 200 mA cm^{-2} with C_2H_4 FE remaining stable at ~70% for over 100 h (Fig. 4g). This result corresponds to an estimated energy cost of 203 GJ t^{-1} for C_2H_4 production using Cu/C–COOH electrodes for COR with an H_2/CO ratio of 2/1 (Fig. 4f and Supplementary Fig. 45), representing a 24% reduction compared with the pristine Cu electrode (267 GJ t^{-1}) and performance comparable to best previous systems using pure CO. This energy cost is lower than that of conventional thermocatalytic syngas-to-ethylene conversion routes (Supplementary Table 5).

Postelectrolysis characterization showed that the Cu particle size remained similar to that of the fresh electrode, although the surface appeared rougher (Supplementary Fig. 46). XPS analysis indicated that the –COOH functional groups remained present on the Cu/C–COOH electrode after extended operation (Supplementary Fig. 47). Cross-sectional SEM and EDS mapping further suggested that the carbon fibre layer, microporous layer and catalyst layer of the custom carbon paper remained intact and functional (Supplementary Fig. 48), consistent with the previous study[50].

Discussion

The electrochemical reduction of CO to C_2H_4 offers a carbon- and energy-efficient pathway to ethylene, but its dependence on costly, high-purity CO feedstock limits its industrial viability. In contrast, syngas is the dominant industrial source of CO and is produced globally at quantities roughly 250 times greater than pure CO. For a syngas-based COR system, achieving high SPCE is essential, as the inherent H_2 excess in syngas increases downstream separation costs. However, optimizing SPCE leads to H_2 dilution and insufficient CO supply to the cathode, which in turn decreases C_2H_4 FE.

In this work, we developed a CO-shuttling medium composed of COOH-functionalized carbon materials that selectively captures CO from syngas and enhances the CO adsorption on Cu surface, thereby transporting CO preferentially over H_2 to the catalyst sites. This approach enables a high local CO availability for C_2H_4 electrosynthesis from syngas under high-SPCE conditions. Implementing a modified Cu in a syngas feed with an H_2/CO ratio of 2/1, we achieved a C_2H_4 FE of 72% and a C_{2+} SPCE of 73% at 200 mA cm^{-2}, resulting in a projected C_2H_4 energy cost of 203 GJ t^{-1}. By enabling direct use of CO-containing streams such as syngas—rather than purified CO, which is expensive and not widely accessible—this approach brings ethylene electrosynthesis closer to industrial practice and improves both economic and environmental sustainability. More broadly, the CO-shuttling strategy offers a generalizable approach for upgrading dilute or mixed-gas streams, opening opportunities for future research on electrosynthetic processes that directly interface with low-carbon sources.

Methods

Materials and chemicals

Copper chloride dihydrate ($CuCl_2{\cdot}H_2O$, Caledon Laboratory Chemical), lithium hydroxide (LiOH, Caledon Laboratory Chemical), sodium borohydride ($NaBH_4$, Caledon Laboratory Chemical), anhydrous iron chloride ($FeCl_3$, Caledon Laboratory Chemical), nickel chloride hexahydrate ($NiCl_2{\cdot}6H_2O$, Caledon Laboratory Chemical), 5 wt% Nafion perfluorinated resin solution, methanol (Caledon Laboratory Chemical), sodium hydroxide (NaOH, Caledon Laboratory Chemical), carbon black (acetylene, Alfa Aesar), graphene oxide (Sigma-Aldrich), carbon nanotube (Sigma-Aldrich), carbon paper-based gas diffusion layers (Freudenberg H23C3, Fuel Cell Store), iridium oxide-coated anode (Ir-MMO-coated titanium felt, Magneto Special Anodes), copper (nanopowder, 25 nm, US Research nanomaterials), anion-exchange membranes (Sustainion x37–50 grade 60, Dioxide Materials). All chemicals were used without further purification. All experiments were conducted using Millipore ultrapure water with a resistivity of 18.2 MΩ cm. Chemicals were used directly without further treatment.

Synthesis of the CuO nanosheets

CuO nanosheets were synthesized via a modified hydrothermal method[51]. Initially, 1.70 g of $CuCl_2{\cdot}2H_2O$ and 3.48 g of sodium dodecylbenzenesulfonate ($CH_3(CH_2)_{11}C_6H_4SO_3Na$) were dissolved in 25 ml of deionized water under continuous stirring at 25 °C. Subsequently, 15 ml of 4 M NaOH was gradually added to the solution, leading to the formation of a dark blue mixture. The solution was transferred into a 50-ml Teflon-lined stainless-steel autoclave and subjected to hydrothermal treatment at 120 °C for 24 h. Upon cooling to room temperature, the resulting black precipitate was collected, washed three times with deionized water and ethanol and separated by centrifugation. The dried powder was then sintered in air at 300 °C for 2 h to remove residual organic components.

Synthesis of COOH-functionalized carbon (C–COOH)

The carbon blacks were activated by dispersing 1-g carbon blacks in 100 ml of 9 M nitric acid solution followed with refluxing at 90 °C for 3 h. After overnight drying in oven at 50 °C, the C–COOH is obtained.

Synthesis of OH-functionalized carbon (C–OH)

The carbon blacks were activated by dispersing 0.1-g carbon blacks in 100 ml of aqueous nitric acid solution (70 wt%) followed by refluxing at 80 °C for 1 h. After overnight drying in oven at 50 °C, C–OH is obtained.

Synthesis of C=O-functionalized carbon (C–C=O)

The carbon blacks were activated by dispersing 0.1-g carbon blacks in 100 ml of aqueous nitric acid solution (70 wt%) followed with refluxing at 60 °C for 4 h. After overnight drying in oven at 50 °C, the C–C = O is obtained.

Synthesis of nitrogen-doped carbon (C–N)

The as-prepared active carbon (AC) was mixed with urea with a mass ratio of 1:10 and then heated up in a tube furnace to 800 °C under a gas flow of 100 sccm Ar and maintained for 1 h. The C–N is finally obtained.

Synthesis of phosphoric-doped carbon (C–P)

C–P was prepared by mixing 50 mg of AC with triphenylphosphine at a mass ratio of 1:10 by grinding, with following heating at 800 °C for 1 h under 100 sccm Ar gas flow.

CuO-derived Cu electrode with/without carbon modification

A total of 20 mg of CuO powder and 80 µl of Nafion were dispersed in 3 ml of methanol. The dispersion was further sonicated for 1 h to form a uniform ink. This ink was then spray deposited onto a 3 × 3 cm^2 H23C3 carbon paper substrate, yielding a CuO electrode with a catalyst loading of 1 mg cm^{-2}. The Cu electrode was prepared by in situ electroreducing the CuO electrode under the COR condition. The preparation of C-modified Cu electrode followed a similar process, except that 5 mg of GO, CNT, C, C–N, C–P and C–COOH powder were added to the ink, respectively.

Preparation of Ni–Fe–B anode electrodes

The Ni–Fe–B catalysts were synthesized using a slightly modified version of a previously reported method[52]. $NiCl_2{\cdot}6H_2O$ (332.8 mg) and anhydrous $FeCl_3$ (227.1 mg) were dissolved in 2 ml of deionized water, while a separate solution of $NaBH_4$ (2 ml, 5 M) was prepared. To prevent uncontrolled hydrolysis and potential precipitation, all solutions were cooled in an ice bath for 10 min before mixing. The Ni and Fe precursors were then added dropwise to the $NaBH_4$ solution. The resulting black mixture was washed three times with water and ethanol, followed by centrifugation and vacuum drying.

Materials characterization

The morphologies of samples were recorded by SEM (QUANTA FEG 250). Transmission electron microscopy (TEM) and EDS images were

obtained on a Thermo Fisher Talos F200X featured with four in column SDD Super- X detectors. X-ray diffraction (XRD) patterns were recorded by using MiniFlex 600 with Cu–Kα radiation (λ = 1.54178 Å). XPS measurements were performed in a Thermo Scientific K–Alpha spectrophotometer with the monochromated Al Kα X-ray radiation source. The binding energies were corrected for specimen charging by referencing C 1*s* to 284.6 eV. The specific surface area, pore volume and pore diameter were characterized using a Micromeritics ASAP 2020. Before the analysis, the samples were degassed at 120 °C for 6 h. The crystalline structure of catalysts was examined with Rigaku Miniflex 600 6 G Benchtop powder XRD.

Electrochemical syngas reduction in alkaline media

COR experiments using syngas were performed in a MEA configuration, as outlined in Supplementary Fig. 3. The MEA consisted of a Cu-based cathode, an IrO_2 (or Ni–Fe–B) anode and an anion-exchange membrane (Sustainion X37–50 grade 60). To construct the electrochemical cell, the membrane was positioned between the two electrodes such that the catalyst layers directly contacted the membrane surface. The anode side was enclosed by a 0.02-inch gasket, providing both electrical insulation and mechanical support. The active reaction area was defined by the gasket opening and fixed to 1 cm^2. The assembled MEA was mounted into a Dioxide Materials fuel-cell housing equipped with serpentine flow fields. The bolts were tightened to ensure uniform compression across the MEA. During operation, 1 M LiOH was circulated through the anode compartment at 30 ml min^{-1}, while the cathode received a controlled CO/H_2 gas mixture regulated by a digital mass-flow controller. Galvanostatic operation was used for the electrolysis tests. The current density was increased stepwise—50 mA cm^{-2}, 100 mA cm^{-2}, 200 mA cm^{-2}, 300 mA cm^{-2} and finally 400 mA cm^{-2}—only after the cell voltage reached a steady state at each step. Long-term stability was evaluated at 200 mA cm^{-2}, for which 2 l of 1 M LiOH was prepared to keep the anolyte composition nearly constant over the extended electrolysis period. LSV measurements were performed in a flow-cell setup at a scan rate of 100 mV s^{-1} under either N_2 or H_2 to evaluate the electrochemical response of the system.

Product analysis in COR from syngas

A 1-ml portion of the outlet gas stream—containing COR products together with unconverted CO/H_2—was withdrawn using a Hamilton gas-tight syringe and injected into a PerkinElmer Clarus 590 GC for composition analysis. The GC coupled with a thermal conductivity detector and flame ionization detector was used for this purpose. The chromatograph contains a Molecular Sieve 5 A Capillary Column and a packed Carboxen-1000 Column, with argon serving as the carrier gas. The flow rates of gas in and out of the cell were determined via a bubble column.

Liquid products were analysed by ^{1}H nuclear magnetic resonance (NMR) spectroscopy (600 MHz, Aligent DD2 NMR spectrometer) with water suppression, in which 0.1 ml of the electrolyte was mixed with 0.1 ml of DMSO (internal standard, diluted to 100 ppm (m/m) by D_2O) and 0.4 ml of D_2O.

SPCE calculation

The FE for the gaseous and liquid products were calculated using the following equations:

$$\mathrm{FE_{gas}} = x_i \times v_g \times \frac{z_{p,i}\mathrm{FP}}{\mathrm{RT}} \times \frac{1}{I_{\mathrm{total}}} \times 100\,\%$$

$$\mathrm{FE_{liquid}} = n_i \times \frac{z_{p,i}F}{Q} \times 100\,\%$$

For gas-phase species, x_i denotes the volume fraction of gas product i, v_g is the outlet gas flow rate (sccm), $z_{p,i}$ is the electron requirement per molecule of product i, F is the Faraday constant, P is the ambient pressure (Pa), R is the ideal gas constant, T is the operating temperature and I_{total} is the applied current.

For liquid products, n_i represents the molar amount of the product i collected and Q corresponds to the total change passed during the sampling period.

The SPCE was determined using:

$$\mathrm{SPCE} = \sum_i^n \frac{I_{\mathrm{total}} \times \frac{\mathrm{FE}_i}{z_{r,i}F} \times 24.5 \times 10^3 \left(\frac{\mathrm{cm}^3}{\mathrm{mol}}\right) \times 60 \left(\frac{s}{min}\right)}{v_{\mathrm{CO,input}}}$$

where $z_{r,i}$ is number of electrons consumed per CO molecule for forming product i, $v_{\mathrm{co,input}}$ is the input CO flow rate (sccm).

Operando Raman measurements

Operando Raman spectroscopy was performed in a custom-designed flow cell using 1 M LiOH as the electrolyte[53]. A Renishaw inVia Raman microscope fitted with a ×63 water-immersion lens and a 785-nm excitation source (operated at 0.1% power) was used for data acquisition. Spectra were collected with the Renishaw WiRE 4.4 software, using two consecutive integrations of 10 s each in extended-scan mode. During measurements, syngas was continuously fed into the gas compartment of the cell. All potentials reported for Raman experiments were referenced to an Ag/AgCl (saturated KCl) electrode and no iR correction was applied.

DFT calculations

DFT was applied to investigate the adsorption of CO and H_2 gas molecules on carbon materials with different functional groups and Cu surfaces with and without carbon modification. All DFT computations were performed with the Vienna ab initio Simulation Package (VASP, v.6.4.2)[54–56], using a plane-wave basis and pseudopotential framework[57]. The exchange–correlation effects were treated using the spin-polarized Perdew–Burke–Ernzerhof generalized gradient approximation[57], while electron–ion interactions were modelled through projector-augmented-wave potentials[58,59]. A plane-wave kinetic energy cutoff of 450 eV was applied throughout. Long-range dispersion forces were included via the zero-damping DFT-D3 correction proposed by Grimme et al.[60]. To prevent artificial interactions between repeated slabs, a vacuum layer of 15 Å was introduced along the surface normal (*z*-axis). The solvent effect was taken into account by using the implicit solvent model, which is implemented in VASPsol[61,62]. The Debye length was set to be 3 Å, corresponding to a 1 M concentration of electrolyte. The relative permittivity of the bulk solvent was chosen as that of water (78.4).

Carbon edges were modelled on the basis of 6 × 6 graphene supercell by subtracting 24 atoms, with edge carbon atoms hydrogenated by hydrogen atoms. COOH-modified carbon edge (C_{edge}–COOH) was modelled on the basis of carbon-edge model by adding COOH group on carbon edge. N- and P-modified carbon edges (C_{edge}–N and C_{edge}–P) were also modelled on the basis of carbon-edge model by replacing one edge carbon atom with one N or P atom. C_{edge}-modified Cu was modelled by adding a carbon-edge layer (consisting of 48 atoms) on the top of 6 × 3 Cu(111) surface with four atomic layers. The 3 × 1 × 1 and 6 × 3× 1 Monkhorst–Pack k-point meshes were used for geometry optimization of carbon edges and C_{edge}-modified Cu, respectively.

Carbon surfaces (C_{surf}) were modelled on the basis of a 3 × 3 graphene supercell. N-, P- and COOH-functionalized carbon surfaces were modelled similarly to those functionalized carbon-edge models. C_{surf}-modified Cu was modelled by adding a carbon surface layer on the top of 3 × 3 Cu(111) surface with four atomic layers. The 3 × 3 × 1 Monkhorst–Pack k-point mesh was used for geometry optimization of carbon surfaces and C_{surf}-modified Cu, respectively.

To construct the amorphous carbon models, we implemented a sampling strategy adapted from the approach reported

by Deringer et al.[63]. Using this framework, we examined 50 pristine amorphous carbon structures along with 150 functionalized variants (Supplementary Fig. 30). The procedure began with DFT relaxation of the amorphous carbon configurations. The 50 initial structures were directly taken from the dataset published by Deringer et al.[63] (located in the 'DFT_relaxed_216at' directory of the supplementary files). The distribution of per-atom formation energies obtained after optimization is summarized in Supplementary Table 6. On the basis of this distribution, the five most stable amorphous carbon structures were retained. For each of these five amorphous carbon structures, two *X*-functional groups (*X* = N, P or COOH) were randomly placed on the carbon framework, repeated ten times for each *X*. This procedure generated 150 amorphous *X*-functionalized carbon structures. Their formation energies are provided in Supplementary Tables 7–21. From these, we selected the most stable structure for each combination of functional group and amorphous carbon, yielding a total of 15 stable amorphous *X*-functional carbon structures. Together with the five unfunctionalized amorphous carbons, this gave 20 representative amorphous structures. We first calculated the CO adsorption energy on these amorphous carbon structures. We then constructed Cu surfaces modified by both the amorphous carbon surfaces (20 structures in total) and edge structures (20 structures in total), and calculated CO adsorption energies on these hybrid interfaces.

Multiphysics modelling

Poisson–Boltzmann was applied to calculate the distribution CO at the electrode–electrolyte interface during COR from CO or syngas, which is implemented in COMSOL (COMSOL Multiphysics v.6.2)[64]. See Supplementary Note 1 for simulation details.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data supporting the findings of this study are available within the paper and Supplementary Information. Source data are provided with this paper.

References

1. De Luna, P. et al. What would it take for renewably powered electrosynthesis to displace petrochemical processes?. *Science* **364**, eaav3506 (2019).
2. Chen, C. et al. Progress toward commercial application of electrochemical carbon dioxide reduction. *Chem* **4**, 2571–2586 (2018).
3. Zhong, M. et al. Accelerated discovery of CO_2 electrocatalysts using active machine learning. *Nature* **581**, 178–183 (2020).
4. Ozden, A. et al. Cascade CO_2 electroreduction enables efficient carbonate-free production of ethylene. *Joule* **5**, 706–719 (2021).
5. Zhao, Y. et al. Conversion of CO_2 to multicarbon products in strong acid by controlling the catalyst microenvironment. *Nat. Synth.* **2**, 403–412 (2023).
6. Rabinowitz, J. A. et al. The future of low-temperature carbon dioxide electrolysis depends on solving one basic problem. *Nat. Commun.* **11**, 5231 (2020).
7. Jouny, M. et al. Carbon monoxide electroreduction as an emerging platform for carbon utilization. *Nat. Catal.* **2**, 1062–1070 (2019).
8. Dinh, C. T. et al. CO_2 electroreduction to ethylene via hydroxide-mediated copper catalysis at an abrupt interface. *Science* **360**, 783–787 (2018).
9. Kim, D. et al. Acid-stable Cu cluster precatalysts enable high energy and carbon efficiency in CO_2 electroreduction. *J. Am. Chem. Soc.* **146**, 27701–27712 (2024).
10. Chen, Y. et al. Efficient multicarbon formation in acidic CO_2 reduction via tandem electrocatalysis. *Nat. Nanotechnol.* **19**, 311–318 (2024).
11. Fan, M. et al. Cationic-group-functionalized electrocatalysts enable stable acidic CO_2 electrolysis. *Nat. Catal.* **6**, 763–772 (2023).
12. Cao, Y. et al. Surface hydroxide promotes CO_2 electrolysis to ethylene in acidic conditions. *Nat. Commun.* **14**, 2387 (2023).
13. Huang, J. E. et al. CO_2 electrolysis to multicarbon products in strong acid. *Science* **372**, 1074–1078 (2021).
14. Xie, K. et al. Bipolar membrane electrolyzers enable high single-pass CO_2 electroreduction to multicarbon products. *Nat. Commun.* **13**, 3609 (2022).
15. Liang, Y. et al. Efficient ethylene electrosynthesis through C–O cleavage promoted by water dissociation. *Nat. Synth.* **3**, 1104–1112 (2024).
16. Esmaeilirad, M. et al. Imidazolium-functionalized Mo3P nanoparticles with an ionomer coating for electrocatalytic reduction of CO_2 to propane. *Nat. Energy* **8**, 891–900 (2023).
17. Yan, J. et al. High-efficiency intermediate temperature solid oxide electrolyzer cells for the conversion of carbon dioxide to fuels. *J. Power Sources* **252**, 79–84 (2014).
18. Deka, D. J. et al. Coke formation during high-temperature CO_2 electrolysis over AFeO3 (A = La/Sr) cathode: effect of A-site metal segregation. *Appl. Catal. B* **283**, 119642 (2021).
19. Kibria, M. G. et al. Electrochemical CO_2 reduction into chemical feedstocks: from mechanistic electrocatalysis models to system design. *Adv. Mater.* **31**, e1807166 (2019).
20. Carbon monoxide market size, share & industry analysis, by application (metal fabrication, chemicals, ore processing and extraction, pharma & biotechnology, and electronics) and regional forecast, 2024–2032. *Fortune Business Insights* https://www.fortunebusinessinsights.com/carbon-monoxide-market-105343 (accessed 26 November 2022).
21. O'Brien, C. P. et al. CO_2 electrolyzers. *Chem. Rev.* **124**, 3648–3693 (2024).
22. Syngas market size, share & industry analysis, by technology (gasification and steam methane reforming (SMR)), by feed (natural gas, coal, petcoke, biomass, and others), by product (ammonia, methanol, hydrogen, liquid fuels, DRI, SNG, electricity, and others), and regional forecast, 2025–2032. *Fortune Business Insights* https://www.fortunebusinessinsights.com/syngas-market-109820 (accessed 28 June 2023).
23. Schwarze, K. et al. Demonstration and scale-up of high-temperature electrolysis systems. *Fuel Cells* **23**, 492–500 (2023).
24. Rao, M. et al. Durability of solid oxide electrolysis stack under dynamic load cycling for syngas production. *J. Power Sources* **451**, 227781 (2020).
25. Xiao, Y. C. et al. Reactive capture of CO_2 via amino acid. *Nat. Commun.* **15**, 7849 (2024).
26. Rochelle, G. T. Amine scrubbing for CO_2 capture. *Science* **325**, 1652–1654 (2009).
27. Hauch, A. et al. Recent advances in solid oxide cell technology for electrolysis. *Science* **370**, eaba6118 (2020).
28. Li, Y. et al. Hydrogen/syngas production from different types of waste plastics using a sacrificial tire char catalyst via pyrolysis-catalytic steam reforming. *Energy and Fuels* **37**, 6661–6673 (2023).
29. Lopez, G. et al. Thermochemical routes for the valorization of waste polyolefinic plastics to produce fuels and chemicals. A review. *Renew. Sustain. Energy Rev.* **73**, 346–368 (2017).
30. Jouny, M. et al. General techno-economic analysis of CO_2 electrolysis systems. *Ind. Eng. Chem. Res.* **57**, 2165–2177 (2018).
31. Bolívar Caballero, J. J. et al. Reforming processes for syngas production: a mini-review on the current status, challenges, and prospects for biomass conversion to fuels. *Appl. Energy Combust. Sci.* **10**, 100064 (2022).

32. Cao, Y. et al. Synthesis gas production with an adjustable H_2/CO ratio through the coal gasification process: effects of coal ranks and methane addition. *Energy and Fuels* **22**, 1720–1730 (2008).
33. Ross, M. B. et al. Designing materials for electrochemical carbon dioxide recycling. *Nat. Catal.* **2**, 648–658 (2019).
34. Dattila, F. et al. Active and selective ensembles in oxide-derived copper catalysts for CO_2 reduction. *ACS Energy Lett.* **5**, 3176–3184 (2020).
35. Verdaguer-Casadevall, A. et al. Probing the active surface sites for CO reduction on oxide-derived copper electrocatalysts. *J. Am. Chem. Soc.* **137**, 9808–9811 (2015).
36. Li, J. et al. Constraining CO coverage on copper promotes high-efficiency ethylene electroproduction. *Nat. Catal.* **2**, 1124–1131 (2019).
37. Dry, M. E. The Fischer–Tropsch process: 1950–2000. *Catal. Today* **71**, 227–241 (2002).
38. Gao, D. et al. Rational catalyst and electrolyte design for CO_2 electroreduction towards multicarbon products. *Nat. Catal.* **2**, 198–210 (2019).
39. Papangelakis, P. et al. Improving the SO_2 tolerance of CO_2 reduction electrocatalysts using a polymer/catalyst/ionomer heterojunction design. *Nat. Energy* **9**, 1011–1020 (2024).
40. Shayesteh Zeraati, A. et al. Carbon- and energy-efficient ethanol electrosynthesis via interfacial cation enrichment. *Nat. Synth.* **4**, 75–83 (2024).
41. Wen, B. et al. Recent progress on porous carbons for carbon capture. *Langmuir* **40**, 8327–8351 (2024).
42. Zhang, Z. et al. Rational design of tailored porous carbon-based materials for CO_2 capture. *J. Mater. Chem. A* **7**, 20985–21003 (2019).
43. Gunathunge, C. M. et al. Surface-adsorbed CO as an infrared probe of electrocatalytic interfaces. *ACS Catal.* **10**, 11700–11711 (2020).
44. Wang, Y. et al. Tunable-quaternary (N, S, O, P)-doped porous carbon microspheres with ultramicropores for CO_2 capture. *Appl. Surf. Sci.* **507**, 145130 (2020).
45. Manmuanpom, N. et al. Enhanced CO_2 capturing over ultra-microporous carbon with nitrogen-active species prepared using one-step carbonization of polybenzoxazine for a sustainable environment. *J. Environ. Manag.* **223**, 779–786 (2018).
46. Lee, B.-H. et al. Supramolecular tuning of supported metal phthalocyanine catalysts for hydrogen peroxide electrosynthesis. *Nat. Catal.* **6**, 234–243 (2023).
47. Egbert, J. D. et al. Counting surface redox sites in carbon-supported electrocatalysts by cathodic stripping of O deposited from N_2O. *J. Catal.* **365**, 405–410 (2018).
48. Herzog, A. et al. Operando Raman spectroscopy uncovers hydroxide and CO species enhance ethanol selectivity during pulsed CO_2 electroreduction. *Nat. Commun.* **15**, 3986 (2024).
49. Zhan, C. et al. Revealing the CO coverage-driven C–C coupling mechanism for electrochemical CO_2 reduction on Cu_2O nanocubes via Operando Raman spectroscopy. *ACS Catal.* **11**, 7694–7701 (2021).
50. O'Brien, C. P. et al. Scalability and stability in CO_2 reduction via tomography-guided system design. *Joule* **8**, 2903–2919 (2024).
51. Li, Z. et al. Room-temperature high-performance H_2S sensor based on porous CuO nanosheets prepared by hydrothermal method. *ACS Appl. Mater. Interfaces* **8**, 20962–20968 (2016).
52. Wang, N. et al. Boride-derived oxygen-evolution catalysts. *Nat. Commun.* **12**, 6089 (2021).
53. Li, F. et al. Molecular tuning of CO_2-to-ethylene conversion. *Nature* **577**, 509–513 (2020).
54. Kresse, G. et al. Ab initio molecular dynamics for liquid metals. *Phys. Rev. B* **47**, 558–561 (1993).
55. Kresse, G. et al. Efficiency of ab-initio total energy calculations for metals and semiconductors using a plane-wave basis set. *Comput. Mater. Sci.* **6**, 15–50 (1996).
56. Kresse, G. et al. Efficient iterative schemes for ab initio total-energy calculations using a plane-wave basis set. *Phys. Rev. B* **54**, 11169–11186 (1996).
57. Payne, M. C. et al. Iterative minimization techniques for ab initio total-energy calculations: molecular dynamics and conjugate gradients. *Rev. Mod. Phys.* **64**, 1045–1097 (1992).
58. Blöchl, P. E. Projector augmented-wave method. *Phys. Rev. B* **50**, 17953–17979 (1994).
59. Joubert, D. From ultrasoft pseudopotentials to the projector augmented-wave method. *Phys. Rev. B* **59**, 1758–1775 (1999).
60. Grimme, S. et al. A consistent and accurate ab initio parametrization of density functional dispersion correction (DFT-D) for the 94 elements H-Pu. *J. Chem. Phys.* **132**, 154104 (2010).
61. Mathew, K. et al. Implicit solvation model for density-functional study of nanocrystal surfaces and reaction pathways. *J. Chem. Phys.* **140**, 084106 (2014).
62. Mathew, K. et al. Implicit self-consistent electrolyte model in plane-wave density-functional theory. *J. Chem. Phys.* **151**, 234101 (2019).
63. Deringer, V. L. et al. Computational surface chemistry of tetrahedral amorphous carbon by combining machine learning and density functional theory. *Chem. Mater.* **30**, 7438–7445 (2018).
64. Li, F. et al. Simulating electric field and current density in nanostructured electrocatalysts. *Phys. Chem. Chem. Phys.* **24**, 25695–25719 (2022).

Acknowledgements

We acknowledge support from the Natural Sciences and Engineering Research Council (NSERC) of Canada and TotalEnergies OneTech Belgium. Support from Canada Research Chairs Program (CRC-2021-00316) is also gratefully acknowledged. Infrastructure provided through the Canada Foundation for Innovation and the Ontario Research Fund supported this work. Computations were performed on the Niagara supercomputer at the SciNet HPC Consortium. SciNet is funded by Innovation, Science and Economic Development Canada; the Digital Research Alliance of Canada; the Ontario Research Fund: Research Excellence; and the University of Toronto.

Author contributions

D.S., R.K.M. and Z.W. supervised the work. F.L. conceived the idea. F.L. designed and carried out all the electrochemical experiments. F.L. designed and carried out all the DFT and multiphysics simulations. F.L. and R.K.M. prepared the paper. R.K.M. synthesized the Ni–Fe–B anode material and performed techno-economic analysis. Z.G. synthesized functionalized carbon materials and performed XPS and SEM characterizations. Y.Y. performed Operando Raman and XRD characterization. Q.W. assisted in XPS, SEM, XRD and Operando Raman characterization. Y.L. performed TEM and EDS characterization. L.D. assisted in the DFT and multiphysics simulations. L.B. conducted the literature survey summarized in Supplementary Table 5. Z.A.D., M.L., Y.C.X., Z.Z., J.L., A.S., Y.G. and Q.S. assisted with experiments. H.G.A. assisted in DFT simulations. M.F., J.E.H., S.P., S.M.H., Y.Z., Y.X. and C.-T.D. contributed to data analysis and discussions. All authors discussed the results and approved the final version.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41893-025-01764-w.

Correspondence and requests for materials should be addressed to Ziyun Wang, Rui Kai Miao or David Sinton.

Peer review information *Nature Sustainability* thanks Ye Wang and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

© The Author(s), under exclusive licence to Springer Nature Limited 2026

nature portfolio

Corresponding author(s): David Sinton

Last updated by author(s): Dec 11, 2025

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our Editorial Policies and the Editorial Policy Checklist.

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed	
☐	☒	The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒	A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☒	☐	The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☒	☐	A description of all covariates tested
☒	☐	A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒	A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☒	☐	For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐	Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	VASP (version 6.4.2) and COMSOL Multiphysics® (version 6.2) were used for simulation in this study.
Data analysis	VASP (version 6.4.2) and COMSOL Multiphysics® (version 6.2) were used for simulation in this study.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

All data are available within the paper and Supplementary Information.

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	Reporting on sex and gender was not relevant to the study.
Reporting on race, ethnicity, or other socially relevant groupings	Reporting on race, ethnicity, or other socially relevant groupings was not relevant to the study.
Population characteristics	Population characteristics were not relevant to the study.
Recruitment	Recruitment was not relevant to the study.
Ethics oversight	Ethics oversight was not relevant to the study.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☐ Life sciences ☐ Behavioural & social sciences ☒ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Ecological, evolutionary & environmental sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	Electrosynthesis of ethylene directly from syngas.
Research sample	Electrode source: Carbon black (acetylene, Alfa Aesar), Graphene oxide (Sigma-Aldrich), carbon nanotube (Sigma-Aldrich), Carbon paper-based gas diffusion layers (GDLs, Freudenberg H23C3, Fuel Cell Store), Iridium oxide coated anode (Ir-MMO coated titanium felt, Magneto Special Anodes), Copper (nanopowder, 25 nm, US Research nanomaterials), Anion exchange membranes (Sustainion x37-50-grade-60, Dioxide Materials), 5 wt.% Nafion® perfluorinated resin solution.
Sampling strategy	Electrochemical measurement: COR experiments using syngas were performed in a membrane-electrode-assembly (MEA) configuration. The MEA consisted of a Cu-based cathode, an IrO2 (or NiFe–B) anode, and an anion-exchange membrane (Sustainion X37-50, grade 60). To construct the electrochemical cell, the membrane was positioned between the two electrodes such that the catalyst layers directly contacted the membrane surface. The anode side was enclosed by a 0.02-inch gasket, providing both electrical insulation and mechanical support. The active reaction area was defined by the gasket opening and fixed to 1 cm2. The assembled MEA was mounted into a Dioxide Materials fuel-cell housing equipped with serpentine flow fields. The bolts were tightened to ensure uniform compression across the MEA. During operation, 1 M LiOH was circulated through the anode compartment at 30 mL min–1, while the cathode received a controlled CO/H_2 gas mixture regulated by a digital mass-flow controller. Galvanostatic operation was used for the electrolysis tests. The current density was increased stepwise—50, 100, 200, 300, and finally 400 mA cm–2—only after the cell voltage reached a steady state at each step. Long-term stability was evaluated at 200 mA cm–2, for which 2 L of 1 M LiOH was prepared to keep the anolyte composition nearly constant over the extended electrolysis period. Linear sweep voltammetry (LSV) measurements were performed in a flow-cell setup at a scan rate of 100 mV s–1 under either N2 or H2 to evaluate the electrochemical response of the system.
Data collection	All data were collected from related instruments by author.
Timing and spatial scale	Start date: Aug.1, 2023; End date: Dec. 1, 2024
Data exclusions	No data were excluded from the analysis.
Reproducibility	Repeated experiments were performed to confirm the reproducibility of experimental findings, and all attempts to repeat the experiment were successful.
Randomization	All samples were allocated into groups to the design of related experiments.

Blinding	Building was not relevant to the study.

Did the study involve field work? ☐ Yes ☒ No

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Plants

Seed stocks	N/A
Novel plant genotypes	N/A
Authentication	N/A